*Artificial Life* Manuscript Submission

# Nonlinear Wavefield Dynamics as a Computational Substrate: Phase-to-Amplitude Conversion and Noise-Tolerant Information Processing in the Nonlinear Schrödinger Equation

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**Abstract.** Identifying the minimal physical conditions under which a substrate can robustly process information is a central question in Artificial Life and unconventional computing. Here we investigate the computational properties of nonlinear Schrödinger equation (NLSE) wavefields by constructing a classification task where information is encoded exclusively in phase gradients, rendering it inaccessible to any static amplitude readout. We first establish a 260-dimensional feature vector extracted from the initial wavefield, which is identical in dimensionality to the evolved-state readout; we find it performs at chance level across all noise conditions, confirming that dynamical evolution is necessary for the phase-encoded information to become observable. We then compare the focusing NLSE against its linear counterpart (LSE) across a sweep of phase jitter. Both achieve near-perfect accuracy at low noise, but diverge as jitter increases: beyond a crossover threshold, the NLSE achieves statistically significant superiority (Cohen's $d \approx 2.8$), while the LSE decays more than twice as fast. A parameter sweep over nonlinearity strength $g$ reveals threshold-driven acceleration ($d^2\text{Acc}/dg^2 > 0$), identifying the onset of a regime where nonlinear self-interaction preserves information that linear dispersion destroys. These findings demonstrate that the NLSE wavefield performs a specific computational operation: converting fragile phase structure into robust amplitude morphology via interference. In particular, we find that the nonlinear term protects this conversion under noise. We discuss implications for the Artificial Life goal of identifying minimal, noise-tolerant physical computing substrates.

**Keywords:** nonlinear Schrödinger equation, computational substrate, noise tolerance, phase-to-amplitude conversion, Artificial Life, nonlinear dynamics

# 1 Introduction

A fundamental question in physical computation is what dynamical properties a substrate must possess to robustly process information under noisy conditions. Digital computation relies on error correction, while biological and physical systems achieve noise tolerance through substrate-level dynamics including wave interference, nonlinear self-interaction, and conservation laws. These substrate dynamics are intrinsic to their respective media rather than imposed by external architecture (Lee et al., 2024; Tanaka et al., 2019). Identifying the minimal physical ingredients for such robustness is both a theoretical question and a practical design challenge for unconventional computing (Jaeger & Haas, 2004; Stepney, 2008; Wright et al., 2022), and a principle central to Artificial Life research, which seeks to identify the minimal physical conditions under which lifelike computation can emerge and function (Bedau, 2003; Gershenson, 2025).

This challenge has a natural parallel in biological systems, where cells routinely face the analogous challenge of distinguishing temporally ordered signals in noisy environments, including ion channel sequences, morphogen gradients, and bioelectric patterns. In this class of system, time-averaged levels are insufficient to determine downstream fate (Briscoe & Small, 2015; Levin, 2021; Purvis & Lahav, 2013).

Nonlinear dispersive wave systems offer a tractable setting in which to study such dynamics: they naturally generate rich spatiotemporal structure from simple initial conditions, and the nonlinear Schrödinger equation (NLSE) (Ablowitz & Segur, 1981; Zakharov & Shabat, 1972), for instance, governs light propagation in Kerr-nonlinear media (Kivshar & Agrawal, 2003), and supports solitons and soliton-like structures (Filippov, 2010) whose interactions can be sensitive to phase, amplitude, and spatial arrangement.

Recent work has established that wave systems governed by the focusing nonlinear Schrödinger equation (NLSE) with nonlinearity strength $g = 1$ can serve as computational substrates, both in theory (Marcucci et al., 2020) and in physical implementations via hydrodynamic

substrates (Marcucci et al., 2023; Pierro et al., 2023) as well as discrete oscillator networks (Borlenghi et al., 2018; Silva et al., 2021).

While nonlinearity is widely recognized as a prerequisite for rich dynamics in physical computing substrates (Lee et al., 2024), the conditions under which nonlinearity itself becomes functionally beneficial for robustness have not been isolated by direct comparison to an appropriate linear baseline. This work addresses a fundamental question: what specific computational advantages do NLSE dynamics provide over linear dispersive dynamics, and under what conditions do these advantages emerge?

To investigate, we construct a classification task where conventional static methods fail by design. We then directly compare the performance of a focusing NLSE system to a dispersive linear Schrödinger Equation (LSE, i.e., $g = 0$; we use this notation throughout to emphasize the direct comparison) system across a sweep of increasing input phase jitter ($\eta$). The observation that the LSE system achieves perfect accuracy for low jitter ($\eta \leq 1.0$) confirms that the task is solvable via linear, time-dependent feature expansion. The central challenge, and the focus of this work, is computational stability under noise.

Prior NLSE-based computation has demonstrated universality and physical implementations, but has not isolated the functional advantage of nonlinearity via direct comparison to linear dynamics. Our contribution:

- A 260-dimensional static control confirms that the classification task is unsolvable without dynamical evolution, even at full feature dimensionality
- First direct NLSE vs. LSE comparison under controlled noise, isolating the functional role of the nonlinear term
- Quantification of the robustness margin: the LSE decays 122.4% faster than the NLSE in the high-noise regime
- Identification of a threshold-driven self-stabilization mechanism via curvature anal-

ysis of the nonlinearity parameter $g$

These findings demonstrate how substrate physics can provide computational robustness without architectural complexity. We ask a specific question: under what conditions do interference-driven nonlinear wavefields provide noise-tolerant information processing, as opposed to merely amplifying signals? By directly comparing NLSE and LSE systems under controlled noise, and showing a clear threshold at which nonlinear self-stabilization becomes beneficial, we identify a concrete computational advantage of a particular physical substrate. This transition threshold marks the onset of a regime we designate **Threshold-Driven Nonlinear Self-Stabilization** (TDNS), in which the nonlinear term preserves separability after the linear dynamics have failed.

This paper is structured as follows: We begin by defining a baseline static classifier (a linear classifier trained on a small set of global descriptors of the initial wavefield, with no access to features generated by dynamical evolution; see Section 3.1 for details) that operates on initial, low-dimensional features and demonstrate that it achieves modest accuracy across the jitter sweep. We then show that while the LSE system experiences significant performance degradation as noise increases, the NLSE system maintains a decisive advantage in the extreme-jitter regime ($\eta \geq 1.73$). Notably, at moderate noise levels, the LSE slightly outperforms the NLSE, revealing a computational cost to nonlinearity that is only recovered when noise severity triggers self-stabilizing dynamics. Quantitatively, the LSE exhibits a performance decay rate that is faster than the NLSE, with the NLSE achieving a peak superiority margin at later times.

The findings of this work establish that nonlinear dynamics (measurable as interference patterns instantiated by phase-gradient driven flow fields) are necessary for computational robustness in noisy environments. Importantly, our results do not imply that nonlinearity is universally necessary, but rather that it becomes functionally effective only once the noise regime exceeds the point where linear dynamics no longer maintain stable representations.

We demonstrate the underlying mechanism is TDNS, observable as **nonlinear self-stabilization occurring past a threshold**—a phenomenon where, beyond a critical nonlinearity, the accuracy as a function of the nonlinearity (represented by parameter weight $g$) exhibits positive curvature. This is interpreted as the nonlinear wavefield interference structures spontaneously converting noisy self-interactions into nonlinear dynamics that resist further degradation—a form of "computational self-focusing" intrinsic to the physics of the stable amplitude structures. We show that this property grants the NLSE access to a specific, high-noise regime that is inaccessible to the linear dynamic baseline.

Prior work on NLSE-based computation has demonstrated universality (Marcucci et al., 2020) and physical implementations (Borlenghi et al., 2018; Marcucci et al., 2023; Pierro et al., 2023; Silva et al., 2021); our contribution is to isolate the functional role of nonlinearity via controlled comparison to a linear baseline.

# 2 Model and Wavefield Dynamics

## 2.1 Nonlinear Schrödinger dynamics

We consider a dimensionless focusing nonlinear Schrödinger equation on a two-dimensional square domain,

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2}\nabla^2\psi - g|\psi|^2\psi, \tag{1}$$

where $\psi(x, y, t) \in \mathbb{C}$ is the complex wavefield, $\nabla^2 = \partial_x^2 + \partial_y^2$ is the Laplacian, and $g > 0$ controls the focusing nonlinearity.

Although the $2D$ focusing NLSE is susceptible to wave collapse (Fibich, 2015), in our parameter regime and integration times the localized structures remain stable while demonstrating soliton-like behavior, which we deem sufficient for computational function and which is visually demonstrated in Figure 1.

In the simulations below we set the nonlinearity variable weight $g = 1$ and work in units where the mass and time scales are dimensionless.

Equation (1) conserves several quantities in the continuous limit, including:

$$M[\psi] = \int |\psi|^2 \, dx \, dy, \quad \text{(mass / norm)} \tag{2}$$

$$H[\psi] = \int \left( \frac{1}{2} |\nabla \psi|^2 - \frac{1}{2} |\psi|^4 \right) dx \, dy, \quad \text{(Hamiltonian)}. \tag{3}$$

We discretize Eq. (1) on a $64 \times 64$ grid using a standard split-step Fourier method (Ablowitz & Segur, 1981) with periodic boundary conditions (see Supplementary Material, Sec. S2). The method alternates linear propagation in Fourier space with nonlinear phase kicks in real space, preserving mass exactly; energy conservation is subject to small splitting error but remains accurate to numerical precision over the integration times used here.

## 2.2 Sech Bump Initial Conditions

To simulate wavefield dynamics we establish soliton-like initial conditions consisting of two localized sech bumps.

For each class $c \in \{0, 1, 2, 3\}$, we construct two localized sech bump centers—let $\mathbf{c}_j = (c_{j,x}, c_{j,y})$ denote the center of the $j$-th bump.

We initialize the field with a superposition of two radially localized sech profiles:

$$\psi_{\mathbf{0}}(x, y) = \sum_{j=1}^{2} \text{sech}\left( \sqrt{(x - c_j^x)^2 + (y - c_j^y)^2} \right), \tag{4}$$

normalized to unit mass and multiplied by a plane-wave phase $\exp(i\mathbf{k} \cdot \mathbf{r})$ to impose a global phase gradient (wavevector). The two sech bumps are then superposed:

$$\psi_0^{(c)} = \psi_{\text{bump,1}} + \psi_{\text{bump,2}}. \tag{5}$$

The classes differ only in the global phase gradient wavevector applied to the same two-bump initial condition. This encodes class information into the initial velocity field, which determines the distinct interference and dynamical patterns generated during NLSE evolution. Encoding class information solely in the phase, while holding the amplitude envelope fixed, provides a stringent test of a physical substrate because the system must convert fragile phase structure into observable spatial morphology during evolution.

### 2.3 Phase jitter as input noise

To test robustness, we inject random phase noise into the superposed initial conditions. After constructing $\psi_0^{(c)}$ as described above, we apply pointwise phase perturbations:

$$\psi_0^{(c,s)}(x,y) = \psi_0^{(c)}(x,y)\ \exp(i\,\delta\Phi(x,y))\,, \tag{6}$$

where $\delta\Phi(x,y)$ is sampled i.i.d. from a uniform distribution in $[-\eta,\eta]$ radians. This form of jitter preserves the gross spatial structure of the initial two-bump field while perturbing its phase gradients, providing a controlled test of robustness to phase-localized distortions. In addition to uniform phase jitter, a small amplitude perturbation (10% of $\eta$, Gaussian) was applied. This perturbation was included to prevent an artificially noise-free amplitude envelope under large phase jitter; it affects both systems identically and does not introduce class-discriminative information.

## 3 Feature Extraction and Classification

We compare two different ways to build feature vectors for classification: a control baseline that uses only simple global descriptors of the initial condition, and an evolved-state readout that uses features derived from the evolved wavefield at time $T$.

We generate $N_{\text{class}} = 100$ samples per class, for a total of $400$ samples in each experiment

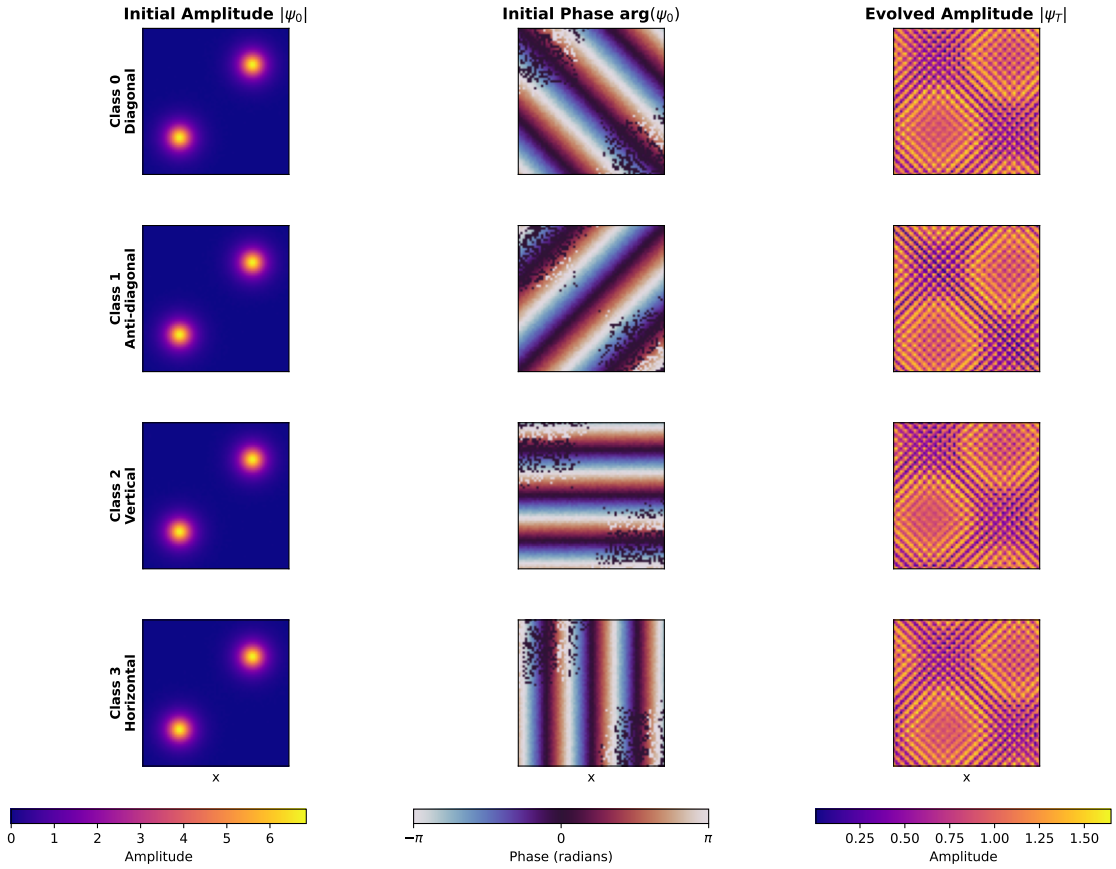


Figure 1: **Sech bump initial conditions encode class information through phase structure:** This figure illustrates representative dynamics on a domain equivalent to those used in quantitative experiments ($L = 20$ with $T = 1.0$ for the main robustness analysis and $T = 2.0$ for the $g$-sweep; see Supplementary Material, Sec. S3). Each row shows one of four classes, differing only applied phase gradient (plane-wave wavevector). **Left:** Initial amplitude $|\psi_0|$ is nearly identical across classes—two localized bumps positioned to generate interference. **Center:** Initial phase arg$(\psi_0)$ encodes class identity through plane-wave momentum imprints (horizontal, vertical, diagonal, anti-diagonal patterns plus jitter). **Right:** Evolved amplitude $|\psi_T|$ after NLSE propagation shows distinct patterns resulting from the phase-dependent interference geometry of the initial conditions. Despite shared initial amplitude profiles, the phase-controlled NLSE wave dynamics produce linearly separable amplitude signatures at time $T$, enabling perfect classification. Jitter level $\eta = 0.1$.

(for each jitter level). A random train–test split with $70$% training and $30$% testing is used throughout.

## 3.1 Control features from the initial condition (Static Baseline)

The control setting is designed to answer the question: How well can a classifier perform if it is only allowed simple, low-dimensional global features of the input, without access to the evolved field? To establish this necessary baseline, we define the **Static Baseline** classifier.

From each jittered initial condition $\psi_0^{(c,s)}$, we construct a $4$-dimensional feature vector:

$$\boldsymbol{\phi}_0 = \big[\langle \cos\varphi \rangle, \langle \sin\varphi \rangle, M[\psi_0], K[\psi_0]\big], \tag{7}$$

where $\varphi(x,y) = \arg(\psi_0(x,y))$ is the phase, $\langle f \rangle$ denotes the spatial average, $M[\psi_0]$ is the total mass (norm), and $K[\psi_0]$ is the total gradient energy. The first two components encode the global phase alignment, while the latter two reflect total energy and mass (see Supplementary Material, Sec. S4 for full definitions).

This feature includes no spatial information and no features of the evolved dynamics. The Static Baseline can only distinguish classes to the extent that they induce differences in these global descriptors, rigorously confirming that the input phase patterns are not linearly separable prior to the system's dynamic evolution.

## 3.2 High-dimensional control from the initial condition

To test whether the evolved system's advantage arises from the dynamics or merely from the higher dimensionality of the feature vector, we extract the *same* 260-dimensional feature vector used for the evolved wavefield (Section 3.3)—256 downsampled amplitude values plus 4 global descriptors—but from the *initial* wavefield $\psi_0^{(c,s)}$ **before any evolution**.

This control matches the evolved-state readout in dimensionality ($\mathbb{R}^{260}$) and classifier (logistic regression with identical hyperparameters), isolating the contribution of the dynamics.

Because the four classes share the same sech-bump amplitude envelope and differ only in phase gradient orientation, the initial amplitude field $|\psi_0|$ is nearly identical across classes. The 260-dimensional $t = 0$ readout therefore has access to the same spatial resolution as the evolved readout but lacks the interference-generated amplitude structure that distinguishes the classes.

Table 1 reports the results across the full jitter sweep. At all noise levels, the 260-dimensional $t = 0$ control performs at or below the 4-dimensional static baseline, reaching $25.8$% at $\eta = 2.0$ (chance $= 25$%). This confirms that (a) the classification task cannot be solved by any static readout regardless of dimensionality, and (b) the computational advantage of the evolved systems is attributable entirely to the dynamics, not to dimensionality expansion.

Table 1: 260-dim $t = 0$ control accuracy (Mean $\pm$ Std., $N = 10$). High-dim initial-state readout performs at/below static baseline.

| $\eta$ | Static (4-d) | $t{=}0$ (260-d) |
|---|---|---|
| 0.10 | .513±.054 | .491±.035 |
| 0.51 | .398±.033 | .394±.043 |
| 1.05 | .358±.047 | .298±.035 |
| 1.59 | .358±.039 | .267±.035 |
| 2.00 | .326±.055 | .258±.030 |

## 3.3 Evolved-state features from the evolved wavefield

In the evolved-state setting, we evolve each initial condition under the NLSE dynamics to the final time $T$ and extract features from the evolved wavefield $\psi_T^{(c,s)}$. Thus the evolved-state features must already be linearly separable for classification to succeed.

Let $A_T(x, y) = |\psi_T(x, y)|$ denote the amplitude field.

To form the readout feature vector, we first downsample the $A_T$ field from its original $64 \times 64$ grid resolution to $16 \times 16$ by subsampling (strided slicing), selecting every fourth pixel in both dimensions. This yields $256$ spatial features. To this, we append the same four global descriptors (mass, gradient energy, and mean sine/cosine of phase), computed at time $T$, forming a final feature vector

$$\boldsymbol{\phi}_T \in \mathbb{R}^{260}. \tag{8}$$

The key point is that $\boldsymbol{\phi}_T$ is a high-dimensional, nonlinear embedding of the initial phase pattern, which is induced entirely by the wavefield dynamics. We train a simple linear classifier (logistic regression) on $\boldsymbol{\phi}_T$, thus leaving the wavefield dynamics and feature mapping fixed to perform the necessary nonlinear transformation.

### 3.4 Classifier and evaluation protocol

For both the Static Baseline and the evolved-state feature sets, we use a multinomial logistic regression classifier with $\ell_2$ regularization, as implemented in `scikit-learn` (Pedregosa et al., 2011). This choice keeps the readout as simple and linear as possible, in accordance with standard practice in physical wavefield computation. The purpose is to ensure that all nonlinear processing capacity is attributed to the wavefield dynamics, not the final classifier.

The dataset for each jitter level consists of $N_{\text{total}} = 400$ samples ($N_{\text{class}} = 100$ per class). We split the data into 280 training and 120 test points (a 70/30 stratified split) and held this split fixed across all runs for reproducibility. The linear readout was calibrated on a fixed subset of evolved states and then evaluated on held-out states generated under independent noise realizations, ensuring that classification performance reflects physical structure produced by the dynamics rather than fitting to noise. Reported accuracies throughout this paper are **test accuracies**, which quantify the generalization performance of the trained classifier on unseen data.

# 4 Experimental Protocol

Both the NLSE and LSE systems operate on features extracted from the evolved wavefield $\psi_T$, isolating the computational effect of the nonlinear term. To explore the differences in computational capabilities we run a series of experiments in which we vary the phase jitter $\eta$ across 15 levels (from $0.10$ to $2.00$), keeping all other NLSE and discretization parameters fixed, with evolution time $T = 1.0$ for the main robustness analysis (parameter sensitivity to $T$ and $g$ is explored in Supplementary Material, Sec. S3).

The jitter strength $\eta$ was varied from $0.10$ (low-noise baseline) to $2.00$ (maximum stress level tested; performance is significantly impaired but above chance) using 15 linearly spaced points. The upper bound of $\eta = 2.00$ was selected to define the physical failure limit of the system, as this value corresponds to a phase uncertainty of approximately $\pm 115^\circ$ (compared to a full $2\pi$ phase cycle).

For each value of $\eta$, the experimental procedure is:

1. For each statistical trial $s \in \{1, \ldots, 10\}$ (seeded independently):
   (a) Generate $N_{\text{class}} = 100$ jittered initial conditions $\psi_0^{(c,s)}$ for each class using the unique random state $s$.
   (b) Compute Static Baseline features $\boldsymbol{\phi}_0^{(c,s)}$.
   (c) Evolve each $\psi_0^{(c,s)}$ independently under LSE ($g = 0$) and NLSE ($g = 1$).
   (d) Compute $\boldsymbol{\phi}_T^{(c,s)}$ evolved-state features.
2. Concatenate the resulting features into three datasets: $X_{\text{Static}} \in \mathbb{R}^{400\times 4}$, $X_{t=0} \in \mathbb{R}^{400\times 260}$ (initial-state control), $X_{\text{LSE}} \in \mathbb{R}^{400\times 260}$, and $X_{\text{NLSE}} \in \mathbb{R}^{400\times 260}$, with corresponding label vector $\boldsymbol{y} \in \{0, 1, 2, 3\}^{400}$.
3. Split the datasets into 280 training and 120 test points (a $70/30$ stratified split).

4. Train a logistic regression classifier on each of the three feature sets and evaluate each on the held-out test set.

5. Record the test accuracies. The final results are reported as the **mean test accuracy** $\mu$ and **standard deviation** $\sigma$ calculated across the $N = 10$ independent statistical trials.

The full sweep covers 15 iterations of phase jitter perturbations from $\eta = 0.10$ to $\eta = 2.00$ (see Table 4).

The 260-dimensional $t = 0$ control (Section 3.2) confirms that the initial amplitude profiles contain no class-discriminative information at any noise level, establishing that the classification task requires dynamical evolution.

# 5 Statistical Methodology

## 5.1 Accuracy comparisons and effect sizes

For each jitter level we ran $N = 10$ statistically independent trials per system (distinct random seeds). To quantify the divergence between the NLSE and LSE systems in the high-noise region, we restricted multiple-comparison correction to the seven highest jitter values, where robustness is of primary interest,

$$\eta \in \{1.1857, 1.3214, 1.4571, 1.5929, 1.7286, 1.8643, 2.0000\}.$$

For each $\eta$ in this set, we computed a two-sample $t$-statistic for the difference in mean test accuracy (NLSE–LSE) using the empirical standard deviations over seeds (degrees of freedom df $= 18$), and obtained two-tailed $p$-values from the Student $t$-distribution. Because NLSE and LSE trials share random seeds, the design is paired; however, we report independent-samples t-tests samples statistics without exploiting this pairing as our goal

is to quantify a robust regime-level separation rather than optimize inference under the paired structure. A paired test would be a natural alternative and may have higher power. Two-tailed tests were chosen because we sought to detect significant differences in either direction—both NLSE superiority at extreme noise and potential LSE advantages at moderate noise. This bi-directional approach helped to enable identification of the crossover threshold where the performance relationship between the NLSE and LSE reverses.

To control the family-wise error rate across these seven planned comparisons in the high-noise regime, we used a Bonferroni correction (Bonferroni, 1936) with a nominal $\alpha = 0.01$, yielding

$$\alpha_{\text{corr}} = \frac{0.01}{7} \approx 0.0014.$$

This conservative choice is appropriate here because the number of pairwise contrasts is small and our primary goal is to avoid false positives when claiming a robustness advantage in the most challenging (high-noise) conditions. Significance was assessed by comparing the raw two-tailed p-values for each contrast to the Bonferroni-adjusted threshold $\alpha_{\text{corr}} = 0.0014$.

Effect sizes are reported as Cohen's $d$ (Cohen, 1988), computed using the pooled standard deviation

$$s_{\text{pooled}} = \sqrt{\frac{\sigma_{\text{NLSE}}^2 + \sigma_{\text{LSE}}^2}{2}},$$

where $\sigma_{\text{NLSE}}$ and $\sigma_{\text{LSE}}$ are the sample standard deviations across the $N = 10$ seeds.

In the extreme-jitter regime ($\eta \geq 1.7286$), the NLSE significantly outperforms the LSE under the Bonferroni-corrected threshold, as shown in Table 2. These correspond to large effects and establish a robustness margin for the nonlinear system at high noise.

For completeness, we note that in the moderate-noise subset the linear system retains an advantage (Table 3).

Together, these results quantify the crossover: a slight linear advantage at moderate noise,

Table 2: Statistical comparison in the high-noise regime ($\eta \geq 1.73$). All $p < \alpha_{\text{corr}} = 0.0014$.

| $\eta$ | $\Delta$Acc | $t(18)$ | $p$ | $d$ |
|---|---|---|---|---|
| 1.73 | +.071 | 4.9 | $1.3\times10^{-4}$ | +2.2 |
| 1.86 | +.111 | 7.3 | $1.0\times10^{-6}$ | +3.2 |
| 2.00 | +.132 | 6.3 | $6.0\times10^{-6}$ | +2.8 |

Table 3: Statistical comparison in the moderate-noise regime ($\alpha_{\text{corr}} = 0.0014$).

| $\eta$ | $\Delta$Acc | $t(18)$ | $p$ | $d$ | Sig. |
|---|---|---|---|---|---|
| 1.19 | $-$.047 | $-$6.1 | $9\times10^{-6}$ | $-$2.7 | * |
| 1.32 | $-$.058 | $-$6.4 | $5\times10^{-6}$ | $-$2.9 | * |
| 1.46 | $-$.040 | $-$3.7 | .0018 | $-$1.6 | – |
| 1.59 | +.005 | 0.4 | .703 | +0.2 | – |

*$p < \alpha_{\text{corr}}$; – not significant

followed by a statistically reliable and substantial nonlinear advantage once the jitter is severe enough to trigger self-stabilizing dynamics where, at the highest jitter level ($\eta = 2.0$), the NLSE's advantage of $+13.17$ percentage points corresponds to a large effect size (Cohen's $d \approx 2.8$), confirming that the robustness margin is both statistically significant and substantial.

## 5.2 Decay rate analysis in the high-noise regime

To complement the comparison, we quantified the overall difference in performance degradation between the NLSE and LSE systems by estimating the decay rate ($\beta$) of their accuracy curves in the critical high-noise regime.

The decay rate $\beta$ is defined as the slope of a linear regression of mean test accuracy ($\mu$) versus jitter strength ($\eta$),

$$\mu(\eta) = \beta\eta + C,$$

fit over the regime

$$1.05 \leq \eta \leq 2.00,$$

corresponding to indices $\eta_{\text{idx}} = 7$ to $14$ (inclusive) in the full sweep, where the LSE performance begins its monotonic decline (the onset of degradation). This yielded

- **LSE system ($g = 0.0$):** $\beta_{\text{LSE}} \approx -0.3903$,
- **NLSE system ($g = 1.0$):** $\beta_{\text{NLSE}} \approx -0.1755$.

We summarize the relative steepness as

$$\text{Relative Steepness} = \left( \frac{\beta_{\text{LSE}}}{\beta_{\text{NLSE}}} - 1 \right) \times 100\% \approx \mathbf{122.4\%},$$

indicating that the LSE accuracy decays approximately $122.4$% faster than the NLSE accuracy in this high-jitter regime. Decay rates were estimated by linear regression on mean test accuracy as a function of jitter $\eta$ over the specified regime. This regression-based measure provides a global, complementary quantification of the NLSE's superior robustness margin beyond the pointwise statistical comparisons.

# 6 Results

## 6.1 Initial Validation: The Nontrivial Classification Task

Before introducing jitter, we validate that the synthetic class definition is not trivially separable by low-dimensional statistics. The four classes are defined to share the same initial amplitude envelope and differ only in phase structure (global phase-gradient orientation). The objective is to confirm that the classification task requires the computational work of the wavefield's dynamics.

At $\eta = 0.1$ (no jitter), the Static Baseline classifier (four global features) achieves a test accuracy of $0.5133$. In contrast, the LSE and NLSE classifiers (utilizing the evolved amplitude field) both achieve $100$% accuracy. The Static Baseline struggles to resolve all four classes, entirely confusing at least one pair. This confirms that (a) the classes are not

trivial to separate using simple global features of the initial condition, and (b) the evolved wavefield does indeed encode the class-specific information in a linearly separable way, as shown in Figure 1; additionally, while the initial amplitude profiles are nearly identical across classes, the phase-driven interference dynamics produce distinct interference patterns that are readily distinguishable by the evolved-state readout.

The 260-dimensional $t = 0$ control, which matches the evolved-state readout in dimensionality but uses initial-state features, performs even worse: $0.491 \pm 0.035$ at $\eta = 0.1$, declining to $0.258 \pm 0.030$ at $\eta = 2.0$. This confirms that the task is unsolvable by any static readout, regardless of feature dimensionality. The computational work is performed entirely by the wave dynamics.

## 6.2 Robustness Sweep: Quantifying the Informational Margin

We varied the phase jitter $\eta$ across 15 levels from $0.10$ to $2.00$ and repeated the training and evaluation protocol for the three feature sets: the static control, the LSE system ($g = 0.0$), and the NLSE system ($g = 1.0$). Table 4 summarizes the resulting test accuracies.

Notably, at low jitter ($\eta \leq 0.78$), the LSE system achieves marginally higher accuracy than the NLSE (e.g., 100.0% vs 99.6% at $\eta = 0.10$). This suggests the nonlinear term introduces a small computational cost when noise levels are insufficient to trigger the self-stabilization mechanism. The NLSE advantage emerges only when the noise regime is severe enough to destabilize linear dynamics.

The results show a performance divergence between the NLSE and LSE systems past the threshold $\eta_{\text{break}} \approx 1.6$. Figure 2 illustrates this margin of superiority. The LSE system moves toward failure, while the NLSE system maintains computational utility deep into the high-noise regime. Specifically, at the extreme jitter limit ($\eta = 2.00$), the NLSE achieves $76.7\% \pm 5.5\%$ accuracy, significantly outperforming the LSE, which achieves $63.5\% \pm 3.6\%$.

It is important to note that as the jitter increases, the LSE system rapidly degrades toward

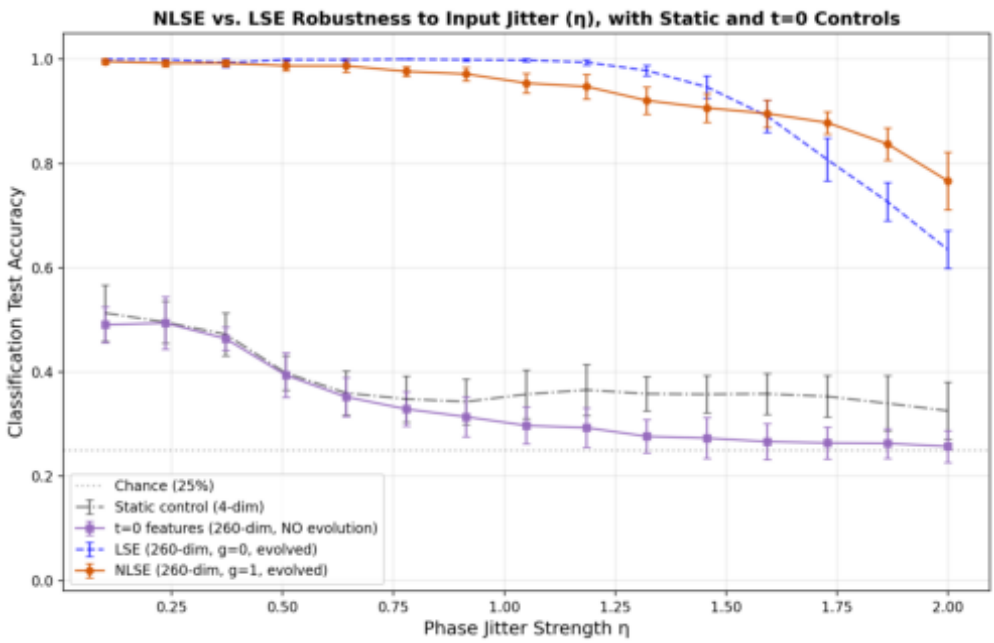


Figure 2: **NLSE vs. LSE Robustness to Input Jitter ($\eta$), with Static and $t = 0$ Controls.** Mean test accuracy ($\mu$) across $N = 10$ seeds; vertical bars show $\sigma$. The 260-dimensional $t = 0$ control (purple) performs at or below the 4-dim static baseline (gray) at all noise levels, confirming that dimensionality alone does not explain the evolved systems' performance. The NLSE (orange) maintains statistically significant superiority over the LSE (blue) in the high-noise regime ($\eta > 1.6$), while the LSE retains a slight advantage at moderate noise.

Table 4: **Test accuracy (Mean $\pm$ Std. Dev.) vs. phase jitter $\eta$.** Results over $N = 10$ seeds. Crossover near $\eta \approx 1.6$: LSE leads at moderate noise before NLSE achieves superiority at $\eta \geq 1.73$.

| $\eta$ | Static | LSE ($g=0$) | NLSE ($g=1$) |
|---|---|---|---|
| 0.10 | .513±.054 | 1.00±.000 | .996±.006 |
| 0.24 | .496±.041 | 1.00±.000 | .993±.007 |
| 0.37 | .473±.041 | .993±.008 | .993±.006 |
| 0.51 | .398±.033 | .999±.003 | .988±.010 |
| 0.64 | .360±.043 | .999±.003 | .988±.013 |
| 0.78 | .348±.044 | 1.00±.000 | .977±.009 |
| 0.91 | .343±.044 | .999±.003 | .972±.013 |
| 1.05 | .358±.047 | .998±.003 | .954±.019 |
| 1.19 | .366±.049 | .994±.005 | .948±.024 |
| 1.32 | .358±.033 | .978±.010 | .921±.026 |
| 1.46 | .358±.036 | .947±.021 | .907±.027 |
| 1.59 | .358±.039 | .891±.031 | .896±.026 |
| 1.73 | .353±.040 | .808±.041 | .878±.021 |
| 1.86 | .340±.053 | .727±.037 | .838±.031 |
| 2.00 | .326±.055 | **.635±.036** | **.767±.055** |

a $63.5$% accuracy level. In contrast, the NLSE system maintains a consistent performance advantage, particularly in the range $\eta > 1.6$, demonstrating its ability to sustain computational utility by actively maintaining class separability above this baseline.

To quantify the functional advantage of the nonlinearity, we analyzed performance in the extreme-noise region ($\eta \geq 1.73$), where NLSE superiority is statistically significant.

**Peak Margin:** The maximum performance divergence (Max $\Delta$Acc) occurs at the highest tested jitter, $\eta =$ **2.00**, where the NLSE sustains a superiority margin of **13.17** percentage points over the LSE.

**Statistical Significance:** To assess statistical reliability, we computed t-statistics for the difference in mean accuracy between NLSE and LSE with Bonferroni correction for 7 comparisons (adjusted $\alpha = 0.0014$). In the moderate-noise regime ($\eta \approx 1.2$–$1.5$), the LSE maintains a small but significant advantage, indicating that the nonlinear term imposes a computational cost when noise is insufficient to trigger self-stabilization. At extreme jitter ($\eta \geq 1.73$), this reverses: the NLSE achieves significant superiority (all $|t| > 4.8$, $p < 0.001$),

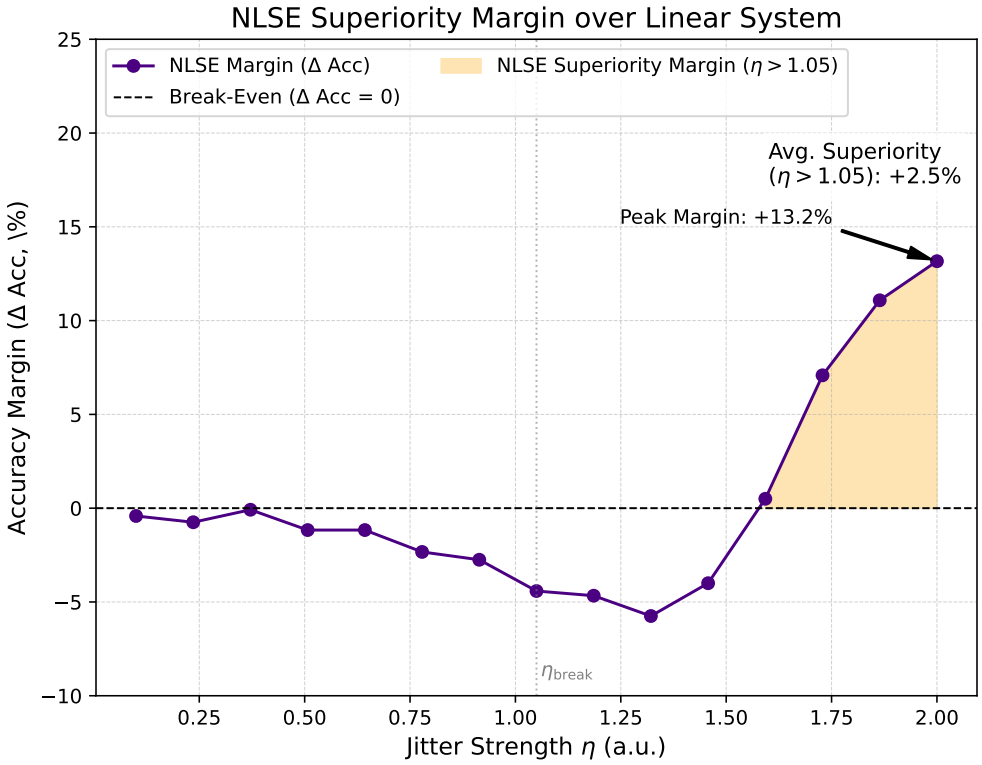


Figure 3: **NLSE Superiority Margin over the Linear System.** The curve displays the difference in mean test accuracy, $\Delta\text{Acc} = \text{Acc}_{\text{NLSE}} - \text{Acc}_{\text{LSE}}$, as a function of phase jitter $\eta$. The gray dashed line marks the threshold crossover $\eta_{\text{cross}} \approx 1.6$ . In the moderate-noise regime ($\eta \approx 1.2$–$1.5$), the LSE maintains a slight advantage. At extreme jitter ($\eta \geq 1.73$), the NLSE achieves decisive superiority, with a peak margin at $\eta = 2.00$.

with Cohen's $d \approx 2.8$ at $\eta = 2.0$. The crossover at $\eta \approx 1.6$ marks the threshold where nonlinear self-stabilization becomes beneficial.

**Utility Threshold:** We define the critical jitter for utility ($\eta_c$) as the maximum noise level at which the mean accuracy remains above **70**%. The NLSE extends computational utility to $\eta_{\mathbf{c}} = \mathbf{2.00}$ units, whereas the LSE's utility collapses earlier at $\eta_{\mathbf{c}} = \mathbf{1.8643}$. The NLSE thus extends the operational regime by **0.1357** $\eta$ units.

**Decay Rate Analysis:** We quantified the robustness by calculating the rate of performance loss across the degradation regime ($\eta \geq 1.05$). A linear regression fit reveals that the LSE system decays at a rate ($\beta_{\text{LSE}} \approx -\mathbf{0.3903}$) that is **122.4**% faster than the NLSE decay rate ($\beta_{\text{NLSE}} \approx -\mathbf{0.1755}$). This is quantitative evidence for the superior noise tolerance of the nonlinear physics.

In summary, the sustained performance advantage of the NLSE, driven by the nonlinear term $g|\psi|^2\psi$, demonstrates that the nonlinear dynamics established by the interference of the initial wavefield components are crucial for maintaining computational stability under high perturbation levels. The fundamental mechanism behind this stability is further explored in Supplementary Material, Sec. S3 through second derivative analysis.

# 7 Discussion

The results establish that the NLSE wavefield performs a specific computational operation: converting phase-encoded information into amplitude-observable structure through wave interference, and protecting that conversion against noise via nonlinear self-interaction. The 260-dimensional $t = 0$ control rules out the alternative hypothesis that the system's advantage arises from dimensionality expansion: initial-state features at full dimensionality perform at chance. This demonstrates that the dynamics are not merely projecting into a higher-dimensional space, but rather that they are creating structure *which does not*

*exist in the initial condition.*

## 7.1 Nonlinearity Provides Computational Robustness

The experiments show that while linear wave dynamics (LSE, $g = 0.0$) can initially separate the phase-encoded classes, the nonlinear dynamics (NLSE, $g = 1.0$) are functionally necessary to preserve that separability under high-noise conditions. The divergence in performance past the crossover $\eta_{\text{cross}} \approx 1.6$ demonstrates that the energy-conserving nonlinear dynamics of the NLSE create a more stable, non-contracting representation of the input.

The crossover behavior observed at $\eta \approx 1.6$—where NLSE transitions from slight deficit to decisive advantage—provides direct evidence that the nonlinear term is not universally beneficial, but rather activates a specific self-stabilization mechanism that becomes necessary only when linear dynamics can no longer maintain feature separability.

The LSE's rapid failure past $\eta_{\text{break}} \approx 1.6$ demonstrates that this regime is indeed too noisy for a purely dispersive wave system to retain feature separability; linear dispersion diffuses the class information, which is quantitatively demonstrated by the LSE's **122.4%** faster decay rate ($\beta_{\text{LSE}} \approx -0.3903$). We refer to the NLSE's capacity to maintain separability in this regime as **Threshold-Driven Nonlinear Self-Stabilization**: once noise exceeds the level where linear dynamics fail, the nonlinear term acts as a computational filter, allowing the system to persistently map noisy input manifolds into a high-dimensional, but non-collapsing, feature space. This property grants the NLSE system access to a computational regime that is closed off to linear wave physics.

The functional superiority of the NLSE in the high-noise regime demonstrates that the nonlinear self-interaction term $g|\psi|^2\psi$ acts as an endogenous stabilization mechanism, continuously correcting dispersive degradation of the class-specific amplitude structures. Furthermore, the positive time derivative ($\text{d}(\text{Acc})/\text{dT} > 0$) confirms this stability is sustained

and beneficial throughout the evolution time $T$, demonstrating that substrate physics alone can provide computational robustness without architectural complexity (Bedau, 2003).

This threshold behavior identifies a regime where the substrate's intrinsic physics determines computational capacity, rather than externally imposed structural-design. The NLSE wavefield requires no engineered error correction, nor training of internal parameters; and likewise, no feedback; i.e., the noise tolerance emerges from the balance between dispersion and self-interaction inherent to the medium. From an Artificial Life perspective, this is significant: the system exemplifies what Bedau (Bedau, 2003) considers organization from the "bottom up", where computational function arises from substrate dynamics rather than imposed architecture, fulfilling a core criterion for identifying minimal, noise-tolerant computational substrates in the Artificial Life program (Gershenson, 2025).

## 7.2 Isolating the Functional Advantage of NLSE Dynamics

Prior NLSE wavefield computation research has focused on three themes: (i) universal approximation and nonlinearity thresholds (Marcucci et al., 2020), (ii) physical implementations in hydrodynamic (Marcucci et al., 2023; Pierro et al., 2023) and discrete oscillator (Borlenghi et al., 2018; Silva et al., 2021) substrates, and (iii) optimization of encoding and readout parameters.

Our contribution is complementary: rather than demonstrating universality or building hardware, we identify what NLSE dynamics compute best: robustness against noise. Our methodological choice centers on functional isolation: By constructing a direct comparison between the NLSE and its linear counterpart (LSE) under increasing phase jitter, we isolate the functional advantage of nonlinear self-focusing dynamics. This intra-system comparison ($g = 1.0$ vs. $g = 0.0$) is critical because it attributes the observed computational stability to the emergent physics of the $g|\psi|^2\psi$ term, rather than confounding factors inherent in engineered systems (e.g., specific network topologies or time-delay feedback

loops). This approach provides a deeper understanding of the medium itself, isolating the functional role of nonlinearity from confounding architectural factors. This isolating of the functional role of nonlinear is a foundational step toward the Artificial Life research goal of identifying self-stabilizing computational substrates that support robust information processing independent of engineered design (Gershenson, 2025; Stepney, 2008).

Specifically, we demonstrate that nonlinear dynamics, established autonomously by interference-driven self-focusing of the localized wavepackets, are necessary to maintain computational utility in the high-noise regime ($\eta > 1.6$). A parameter sweep over $g \in [0, 1]$ (Supplementary Material, Sec. S3) revealed that this robustness is not a continuous, increasing function of the nonlinearity, but is rather governed by a nonlinear self-stabilization mechanism occurring past a threshold. This quantitatively demonstrated by the positive second derivative ($\mathrm{d}^2(\mathrm{Acc})/\mathrm{d}g^2 > 0$) observed in the critical $g$ regime. To our knowledge, prior NLSE wavefield computation work has not explicitly isolated this robustness mechanism, with existing frameworks either targeting function approximation or requiring physically engineered bias elements (e.g., stationary KdV solitons (Marcucci et al., 2023)) or external parameter tuning (Pierro et al., 2023).

We note that our architecture differs from standard reservoir computing in that the NLSE is not driven by a temporal input signal during evolution; rather, class information is encoded in the initial phase structure and the system evolves autonomously. This places our approach closer to Neural ODE frameworks (Chen et al., 2018), where computation is performed by an ODE solver applied to an initial state. The key distinction is that the NLSE's conservation laws and nonlinear self-interaction provide a physically grounded mechanism for noise-tolerant computation that is absent in generic neural ODEs.

The spontaneous transformation of noisy phase geometry into a linearly separable spatial amplitude morphology is the underlying process by which the NLSE provides its margin of superiority. The linear system quickly succumbs to entropic mixing at high jitter, whereas

the NLSE's self-stabilizing nature preserves the class distinctions, a critical requirement for a practical computational medium.

## 7.3 Mechanism: Threshold-Driven Nonlinear Self-Stabilization

The functional regime-superiority of the NLSE system arises from the physics of the substrate itself performing noise-filtering transformation. The robust performance exhibited by the NLSE system in the Critical Regime ($\eta = 2.0$, $T = 2.0$) is achieved through interference driving nonlinear dynamics which lead to self-focusing stabilization past the threshold of ($\eta > 1.6$). The positive mean derivative $d(\text{Acc})/dT = +0.3389$ (from the time sweep) confirms that this $g$-driven structural stability is sustained and beneficial throughout the evolution time $T$, preventing the features from collapsing. The mechanism is, thus, consistent with threshold behavior, not a continuous linear process.

Additionally, as demonstrated by the $g$-sweep (Supplementary Material, Fig. S1), the performance is not a smooth, linear function of $g$. While the overall mean $d^2\text{Acc}/dg^2$ is ($-0.5660$) due to saturation (overall diminishing returns), the curve exhibits a sharp concave-up region in the critical range of $g \in [0.3, 0.7]$. The positive local second derivative, peaking at $+1.2500$ at $g \approx 0.3$, confirms a **threshold-driven acceleration** in accuracy. This is the physical signature of the nonlinear self-stabilization mechanism: the $g$ term acts as a continuous control parameter, driving the wavefield's amplitude structure from a rapidly diffusing state to a stable, self-focused, localized amplitude pattern once the nonlinearity overcomes the critical noise threshold ($\eta > 1.6$).

The linear system, conversely, lacks this mechanism entirely. The LSE fails because its dynamics are constrained to diffusion, erasing the encoded phase gradients over the necessary evolution time $T$. The NLSE succeeds because the nonlinear dynamics compute the required stability, converting disordered phase differences into classification-preserving amplitude patterns–a configuration that is demonstrably stable and improving across the

evolution time, as shown by the positive $d(\text{Acc})/dT$ result.

## 7.4 Why Phase-only Encoding as a Test

Encoding class information purely in the initial phase, while keeping the amplitude envelope fixed, is a stringent test for any physical substrate:

- Many physical observables (e.g. intensity measurements) are primarily sensitive to amplitude. A system that relies only on instantaneous amplitude differences would fail this test.
- In the present system, phase controls interference between the localized sech-profiled wavepackets. The subsequent nonlinear dynamics convert these phase differences into amplitude and gradient structure, which can be observed at time $T$.
- The control classifier, with only four global features, does not see enough of this latent structure to perfectly recover the labels.
- The 260-dimensional $t = 0$ control provides the definitive test: even with access to the full spatial resolution of the initial amplitude field, a linear classifier cannot distinguish the classes. This rules out the possibility that the task is trivially solvable by any static method, confirming that the wave dynamics perform genuine computation.

That the evolved-state readout achieves 100% accuracy for moderate jitter levels and high robustness otherwise therefore provides strong evidence that the nonlinear wavefield is functioning as a genuine computational substrate, not merely as a trivial feature expander.

## 7.5 Robustness, Jitter, and Noise

The robustness sweeps shown in Table 4 demonstrate that the NLSE's self-stabilizing dynamics create a high degree of noise tolerance. This suggests that the class-specific manifolds in the evolved-state space are relatively stable under small perturbations of the

initial condition. Although the NLSE is a conservative Hamiltonian system and formally lacks attractors, the four classes map to distinct, well-separated regions of the evolved-state space, such that jittered initial conditions remain within their respective class boundaries. Of course, in more complex settings and with more realistic noise models, perfect accuracy may not persist. The current results should be viewed as a proof-of-concept in a controlled synthetic environment. Nonetheless, they illustrate that the nonlinear dynamics of wavefields established by self-focusing interference patterns can exhibit precisely the kind of robust, noise-tolerant separation of inputs that wavefield computation seeks to exploit.

## 7.6 Limitations and Future Directions

The present work has several limitations that also point toward natural extensions:

- **Synthetic Task:** The classification task is synthetic and low-dimensional (four classes). An obvious next step is to couple the NLSE system to more realistic inputs, or to embed higher-dimensional data (e.g. images) into phase or amplitude patterns and test more challenging tasks.
- **Multi-time Readouts:** We only use features extracted at a single final time $T$. In many reservoir computing applications, one exploits the full temporal evolution (e.g. by concatenating features across time). Investigating multi-time readouts will be addressed in future work.
- **Comparison to Other Dynamical Systems:** While we included a simple control baseline, we did not compare the NLSE system to other standard dynamical models (e.g. echo state networks or random recurrent networks) on the same synthetic task. Such comparisons would clarify what is unique (or not) about interference-driven nonlinear wavefields.
- **Relationship to Reservoir Computing:** Our system is driven by initial conditions

rather than temporal input signals, and thus does not satisfy the standard definition of reservoir computing. Future work will investigate temporally driven NLSE systems with echo state properties, as suggested by related work on wave-based dynamical systems (Keller et al., 2024).

This work uses synthetic four-class phase patterns as a controlled testbed; extensions to real-world data and temporal processing are natural next steps. More broadly, one could imagine extending this framework to: incorporate dissipation or driving, turning the NLSE into a driven-dissipative system closer to experimental physical substrates; use more biologically or physically motivated initial conditions (e.g. experimental imaging data mapped into phase and amplitude); study memory capacity and temporal processing capabilities by feeding in time-varying inputs rather than one-shot initial conditions.

# 8 Conclusion

This work isolated the functional advantage of nonlinear dynamics in nonlinear wavefield computation. A 260-dimensional control extracted from the initial wavefield—matching the evolved-state readout in dimensionality—performs at chance across all noise levels. This confirms that it is the dynamics of the system which directly produce the computational advantage, as opposed to an advantage conferred by dimensionality. By constructing a direct comparison between the NLSE system ($g = 1.0$) and its linear counterpart (LSE, $g = 0.0$), we established the central thesis: dynamical nonlinearity is a key factor driving computational robustness once the noise level exceeds the linear regime's stability threshold.

The quantitative evidence supporting this functional necessity is strong. In the high-noise regime ($\eta > 1.6$), the NLSE maintains superiority, with statistically significant advantages emerging at extreme noise ($\eta \geq 1.73$). The LSE exhibits a decay rate that is $122.4$% faster than the NLSE, which sustains a peak superiority margin of $13.17$ percentage points. Statis-

tical significance was confirmed via t-tests, with effects at the highest jitter levels ($\eta \geq 1.73$) surviving Bonferroni correction (all $|t| > 4.8$, $p < 0.001$; we used $\alpha = 0.01$ given the small number of planned contrasts). This divergence demonstrates the functional boundary between linear diffusion and nonlinear self-stabilization.

These metrics constitute the quantitative signature of the specific mechanism which we term Threshold-Driven Nonlinear Self-Stabilization (TDNS). An initially recognized implication of TDNS is that there exists a regime of physical dynamics where nonlinear evolution preserves information that linear evolution inevitably destroys. As evidenced by our second derivative analysis ($\mathrm{d}^2(\mathrm{Acc})/\mathrm{d}g^2 > 0$), the system undergoes dynamics consistent with threshold behavior where the balance between dispersion and nonlinearity causes the wavefield to accelerate into stable self-focusing localized amplitude patterns. We have likewise shown that the LSE significantly fails to maintain computational efficacy in this exact regime where the NLSE sees an acceleration followed by a shift in behavior. Taken together, this heavily suggests a nonlinear dynamical mechanism is the physical means by which the system maintains informational computability by actively converting disordered phase distinctions into linearly separable, structured amplitude patterns.

The ability of the NLSE to access this high-noise regime provides a physical example of a system whose computational boundaries are defined by its capacity for self-stabilization, independent of complex architectural design. This finding establishes a crucial conceptual foundation for designing robust physical computing architectures and is highly relevant to understanding emergent, noise-tolerant information flow in dynamic physical systems and biological systems–this being a question of central concern in the field of Artificial Life.

Further, our work provides a concrete, mechanism-level demonstration of substrate-level computation in a continuous physical system. Here, the NLSE system offers a tractable case of a threshold transition from a linear diffusion computational regime to a nonlinear self-stabilization computational regime which can be identified, quantified, and experi-

mentally reproduced. These results point toward potential implementations in photonic, hydrodynamic, and other nonlinear physical substrates where threshold-driven stabilization may be exploited for robust computation.

These results suggest that the design of noise-tolerant physical computing systems should attend not only to the presence of nonlinearity but to the specific balance conditions under which nonlinear self-interaction overcomes entropic degradation—a principle relevant to photonic, hydrodynamic, and other nonlinear substrates where threshold-driven stabilization may be exploited from ubiquitous wave-like dynamics, conservation laws, and general nonlinear self-interaction. More broadly, these results contribute to the Artificial Life research program by demonstrating that a continuous physical system can exhibit a quantifiable transition from linear-diffusive to nonlinear-stabilized computation, offering a concrete example of the kind of minimal, self-organizing computational substrate that the field seeks to identify (Bedau, 2003; Gershenson, 2025).

# Data and Code

Data supporting the findings of this work are available from the author upon reasonable request. Simulation code can be found at:

https://github.com/amr28693/solitonic_reservoir

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